

Xinwei Qiang

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Research interests: GPU systems, compilers, and distributed systems.

Publications

[*NeurIPS '26*] AsymHP: Load-Balanced Sparse Attention for Video Diffusion Transformers.

Xinwei Qiang*, Yue Guan*, Ruihan Zhu, Mihir Jagtap, Zaifeng Pan, Zhongkai Yu, Chang Chen, Zhengding Hu, Yufei Ding, Adnan Aziz

[*ASPLOS '27*] Impeller: Scalable and Efficient Zero-Knowledge Proof Computation Framework via Graph Compilation.

Liukun Yu*, **Xinwei Qiang***, Shixuan Sun, Chen Chen, Jingwen Leng, Jin Dong

[*SC '26*] FluxZK: Scalable and Efficient Zero-Knowledge Proof Computation via GPU Acceleration.

Xinwei Qiang*, Liukun Yu*, Xiyu Wang, Zhengyi Li, Shixuan Sun, Jingwen Leng, Chen Chen, Jiaping Gui, Zhenzhe Zheng, Jin Dong, Minyi Guo

[*ICML '26*] TritonGym: A Benchmark for Agentic LLM Workflows in Triton GPU Code.

Yue Guan*, Yichen Lin*, Xu Zhao, Jianzhu Yao, **Xinwei Qiang**, Zhongkai Yu, Pramod Viswanath, Yufei Ding, Adnan Aziz

[*OSDI '26*] Syncopate: Efficient Multi-GPU AI Kernels via Automatic Chunk-Centric Compute-Communication Overlap.

Xinwei Qiang*, Yue Guan*, Zhengding Hu, Keren Zhou, Yufei Ding, Adnan Aziz

[*ICLR '26*] DASH: Deterministic Attention Scheduling for High-throughput Reproducible LLM Training.

Xinwei Qiang, Hongmin Chen, Shixuan Sun, Jingwen Leng, Xin Liu, Minyi Guo

[*SOSP '25*] Mercury: Unlocking Multi-GPU Operator Optimization for LLMs via Remote Memory Scheduling.

Yue Guan, **Xinwei Qiang**, Zaifeng Pan, Daniels Johnson, Yuanwei Fang, Keren Zhou, Yuke Wang, Wanlu Li, Yufei Ding, Adnan Aziz

* Equal contribution.

Preprints

[*arXiv '26*] Beyond Parallel Blindness: Information Floors and Model Gaps in Block Drafting.

Xinwei Qiang, Xiang Fang, Chang Chen, Zaifeng Pan, Yue Guan, Yufei Ding

[*arXiv '26*] GF-DiT: Scheduling Parallelism for Diffusion Transformer Serving.

Xinwei Qiang, Yifan Hu, Shixuan Sun, Jing Yang, Han Zhao, Chen Chen, Yu Feng, Jingwen Leng, Minyi Guo

Research Appointments

Research Assistant, EPCC Lab, Shanghai Jiao Tong University

Jul 2024 – Jun 2026

Co-advised by Profs. [Jingwen Leng](#) and [Shixuan Sun](#)

Research Intern, Picasso Lab, University of California San Diego

Jul 2025 – Dec 2025

Advisor: Prof. [Yufei Ding](#)

Education

University of California San Diego — Ph.D. in Computer Science

Sept 2026 – Present

Advisor: Prof. [Yufei Ding](#)

Shanghai Jiao Tong University — B.E. in Computer Science

Sept 2022 – Jun 2026

John Hopcroft Class, an honors program in computer science.

Service

Reviewer: ICLR 2027

Technical Skills

Programming: C/C++, CUDA, Python, Rust, Triton